MIL-R-46631B(MU)
8 June 1972
SUPERSEDING
MIL-R-46631A(MU)
9 October 1964

MILITARY SPECIFICATION

REMOVER, AIRCRAFT CANOPY, M1A3, ASSEMBLY

1. SCOPE

1.1 This specification covers Remover, Aircraft Canopy, M1A3 Assembly loaded with Cartridge Aircraft Canopy Remover, M29A2, for use in conjunction with other propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitations for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons; General Specification for

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MS-28741	- Hose Assembly - Detachable End Fitting, Medium Pressure

DRAWINGS

Munitions Command

F8593731	- Remover, Aircraft Canopy, M1A3, Assembly
F8797268	- Carton, Packing, Ammunition for Remover, Aircraft Canopy, M1A3
F8797269	- Box, Packing, Ammunition for Remover, Aircraft Canopy, M1A3

FSC 1377

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2.1 (Cont'd)

IEL-8593731 - Index of Inspection Equipment List for Remover Aircraft Canopy, M1A3 Assembly

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ANSI-B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N. Y. 10018.)

3. REQUIREMENTS

3.1 General.-Material, parts, and assemblies shall comply with the requirements specified on the applicalbe drawings and referenced specifications.

3.2 Torque.-Torque requirements shall be as specified in Table I. No loosening or deformation shall occur upon application of the breakaway torque.

Table I

Components	B Seating or Bottoming (Inch Pounds)	C Breakaway (Inch Pounds)
Firing head and inside tube after inserting cartridge	400 + 50	300

3.3 Assembly.

3.3.1 The boss face surface of the firing head shall be in complete contact with the cartridge.

3.3.2 The firing head assembly shall be free to rotate 360 degrees.

3.4 Residual magnetism.-The remover shall not deflect the indicator of the compass more than 3 degrees in either direction when passed in front of the compass at a distance of 6 inches.

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3.5 Functional.

3.5.1 Ballistics.-Removers loaded with accepted M29A2 cartridges shall propel a 300 pound mass vertically and shall comply with the requirements listed in Table II.

Table II

Temperature	Velocity (Min.) (see 6.4)	Thrust (Min.)	Ignition Delay (Max.) (see 6.3)
-65° to -70°F	17 fps	2000 lbs.	40 milliseconds
65° to 75°F	20 fps	2800 lbs.	40 milliseconds
160° to 165°F	24 fps	3400 lbs.	40 milliseconds

3.5.2 The remover shall not fail to fire.

3.5.3 The remover shall function without mechanical failure (see 6.2).

3.6 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.6.1 Metal defects.-The component shall be free from cracks, splits, cold shuts, inclusions, porosity or similar defects.

3.6.2 Burr.-The component shall not have a burr which might interfere with assembly or function of the item or which might be injuriuos to personnel handling the item.

3.6.3 Foreign matter.-The component or assembly shall not contain chips, dirt, grease, rust corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 to define the Quality Assurance Terms used.

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4.1.2 Inspection.-Inspection shall be in accorance with MIL-A-2550, except as specified herein.

4.2 Inspection provisions.

4.2.1 Lot.

4.2.1.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105.

4.2.1.2 Size of lot.-A lot shall consist of an identifiable group of assembled units containing metal parts assemblies produced by a homogeneous process and loaded with a single lot of accepted cartridges.

4.2.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated critiera in Table III. The total number of allowable defects per class for either the major or minor classification shall not exceed the acceptance number of Table III. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

Table III

Lot Size	Number of Samples	Major		Minor	
		Accept	Reject	Accept	Reject
1-300	100% Examination				
301-800	50	0	1	1	2
801-1800	90	1	2	2	3
1801-10000	200	2	3	3	4

4.2.2.1 Cartridge, Aircraft Canopy Remover, M29A2, Dwg. D8593625; a detail of Dwg. F8593731.

Defect	Method of Inspection
CRITICAL	
1. Mixed cartridges	Visual
2. Metal defective	Visual
3. "O" ring missing or improperly secured	Visual

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4.2.2.1 (Cont'd)

Defect	Method of Inspection
4. Sealing disc missing or damaged	Visual
5. Marking missing, incorrect or illegible	Visual

MAJOR

101. Cartridge discolored, dirty, oily, corroded or stained	Visual
102. Case dented	Visual
103. "O" ring split, cut or cracked	Visual

MINOR

201. Case scratched, wrinkled or bulged	Visual
202. Varnish missing or inadequate	Visual

4.2.2.2 Assembly, Dwg. F8593731, less all drawings except C8593729, D8593792, C8593793, C8593794 and MS24665-302.

CRITICAL

None defined

MAJOR

101. Thread damaged	Visual
102. Burr	Visual
103. Foreign matter	Visual

MINOR

None defined

4.2.2.3 Assembly, Dwg. F8593731, less Dwgs. B8593658, D8593734, C8593786, C8593787 and C8593791.

CRITICAL

1. Boss face surface of firing head not in complete contact with cartridge	Gage
2. Cartridge missing	Visual
3. Seal on inside tube missing	Visual

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4.2.2.3 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined	

4.2.2.4 Assembly, Dwg. F8593731, less Dwg. A8594748.

CRITICAL	
1. Tubes not free to slide	Manual
MAJOR	
101. Firing head not free to rotate	Manual
MINOR	
None defined	

4.2.2.5 Assembly, Dwg. F8593731.

CRITICAL	
1. Mixed removers	Visual
2. Cotter pin ends not spread	Visual
3. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Total length, max.	Gage
102. Locking wire missing or improperly secured	Visual
103. Safety pin missing	Visual
MINOR	
201. Location of labels, incorrect	Visual
202. Car seal missing or insecure	Visual
203. Protective finish inadequate or defective	Visual

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4.2.2.6 Packing in Carton, Dwg. F8797268.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Carton damaged	Visual
203. Packing wet	Visual
204. Sealing strip missing, wrinkled or incomplete	Visual
205. Marking missing, incorrect or illegible	Visual

4.2.2.7 Packing in Wooden, Box, Dwg. F8797269.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Hardware missing, loose, broken or improperly engaged	Visual/Manual
203. Strapping missing, loose, broken or improperly assembled	Visual/Manual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of units incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.2.3 Inspection testing.-The following test shall be perfored in accordance with 4.3.

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4.2.3.1 Breakaway torque.-A breakaway torque shall be applied to each remover threaded joint listed in Table I. Failure of any remover to comply with the requirements of 3.2 shall be cause for rejection of the remover.

4.2.3.2 Residual magnetism.-Each remover shall be tested for excessive residual magnetism. Failure of any remover to comply with the requirements of 3.4 shall be cause for rejection of the remover.

4.2.3.3 Functional.-The following tests shall be performed by the Government.

4.2.3.3.1 Ballistic.-Twenty-one units shall be tested to determine compliance with the requirements of 3.5. The units shall be temperature conditioned as provided in Table IV and 4.3.3.1.1. Failure of any unit to comply with the requirements of 3.5.1, 3.5.2, and 3.5.3 shall be cause for rejection of the lot.

Table IV

Temperature	Number of Samples
-65° to -70°	7
65° to 75°	7
160° to 165°	7

4.2.4 Inspection equipment.-The examination and tests shall be made using equipment listed on IEL-8593731.

4.3 Test methods and procedures.

4.3.1 Breakaway torque.-A torque wrench, set at the torque requirements specified in 3.2, shall be used to determine the adequacy of the joints. The assembly shall be secured in the fixture prior to application of the torque.

4.3.2 Residual magnetism.-The remover shall be tested for excessive residual magnetism with an approved compass, MS 17932 in an area free of local magnetic effects, by placing the remover six (6) inches from and at the same height as the compass, in a north-south horizontal position (firing head assembly at south) with the compass set in an east-west heading. The remover shall be moved slowly past the compass in a longitudinal direction for its entire length, rotated 180 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass.

4.3.3 Functional.

4.3.3.1 Ballistics.-The remover shall be rigidly supported in the fixture by the mounting points and initiated by removing the safety pin. The remover

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4.3.3.1 (Cont'd)

shall comply with the requirements of 3.5. The remover shall be tested within 10 minutes after removal from the conditioning chamber. When this time limit is exceeded, the remover shall be reconditioned at the original temperature.

4.3.3.1.1 Temperature conditioning.-A simulated remover loaded with a dummy Cartridge, containing a temperature sensitive element, shall be placed in a conditioning chamber at the same time as the sample. The conditioning time shall be a minimum of one (1) hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.-Level A. - Packing and marking shall be in accordance with Drawings F8797268 and F8797269.

6. NOTES

6.1 Ordering data.-Procurement documents should specify tht title, number and date of this specification.

6.2 "Mechanical failure" is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.3 "Ignition delay" is defined as the time that elapses between the firing of the cartridge primer and the beginning of the continuous rise on the thrust time record.

6.4 Velocity applies at tube separation.

6.5 Loaded Assembly Hazard Information:
Explosive Hazard Classification: Class 1
Storage Compatiblity: Group B, E, or N
Department of Transporation (DOT) Hazard Class C
DOT Marking - Explosive Power Devices - Class C
Handle Carefully - (Keep Fire Away)

6.6 Hazard notice.-The cartridge described herein and certain of its components are flammable and/or explosive and consequently present hazards in manufacture, handling, storage and shipment. The contractor should recognize these hazards and take appropriate measures to guard and protect against fire, explosion, adverse environment. corrosive atmosphere, rough handling and electrically induced incidents.

Custodian:

Army - MU

Preparing Activity:

Army - MU

Project No. 1377 -A508

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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